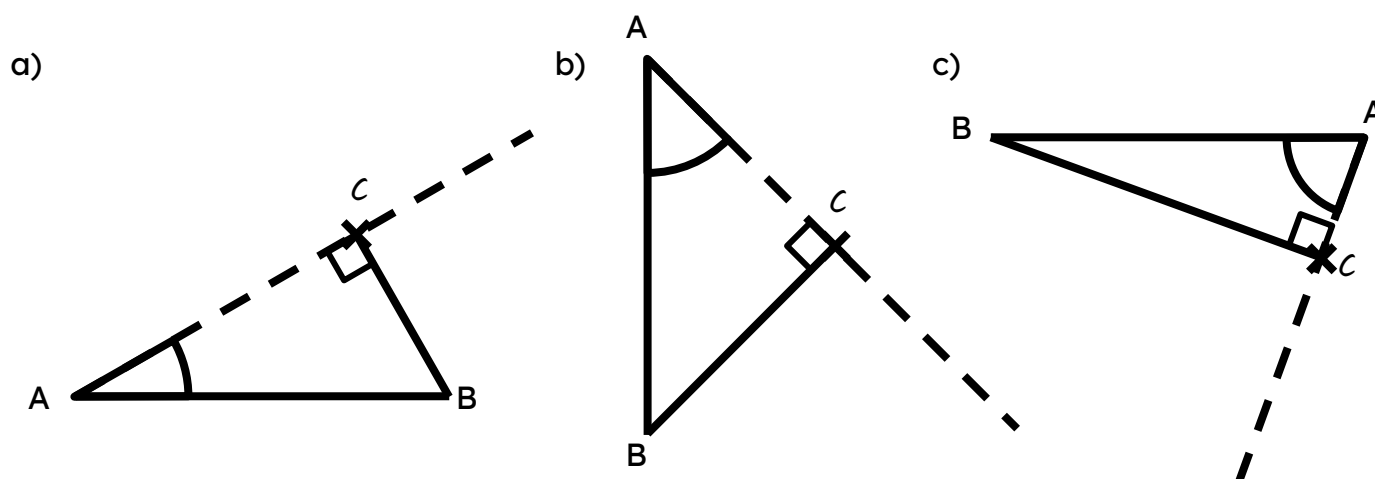




Congruent triangles (RHS)

Task A: Finding the right angle

1) Using a pair of compasses, find and mark the location of vertex C, which makes the minimum length of BC.

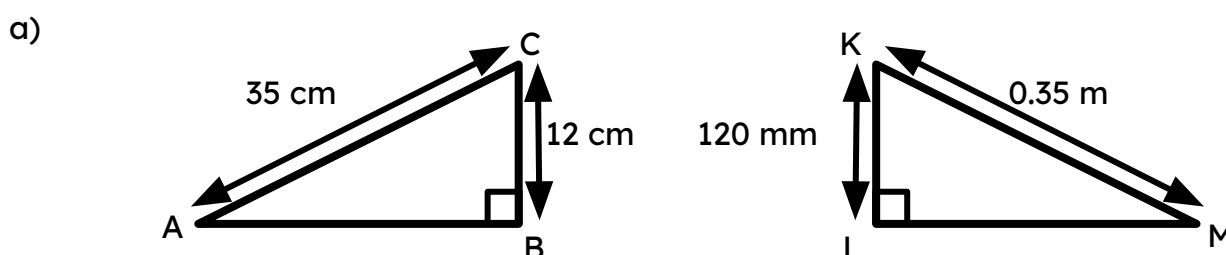


2) Use a protractor to measure angle BCA on each of the diagrams from question 1. What do you notice?

They are all 90°

Task B: Justifying congruence by RHS

1) Use RHS to prove the following pairs of triangles are congruent.



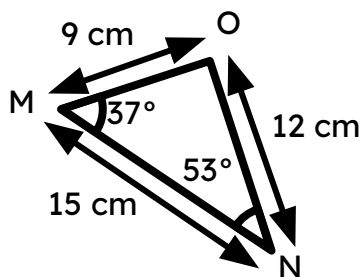
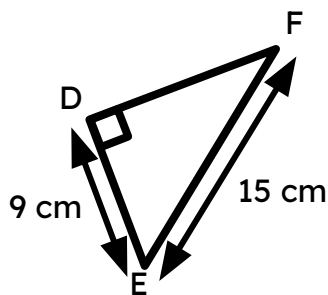
$$\angle ABC = \angle KLM = 90^\circ$$

*AC and KM are each triangle's **hypotenuse** and are equal as $35\text{ cm} = 0.35\text{ m}$*

CB = KL as $12\text{ cm} = 120\text{ mm}$,

So, triangle ABC and KLM are congruent by RHS.

b)



$$\angle MON = 180 - (37 + 53) = 90^\circ = \angle EDF$$

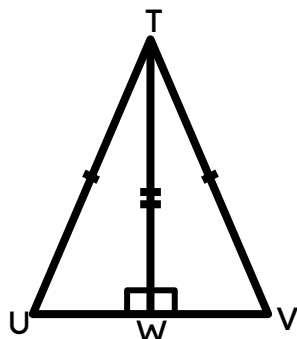
EF and MN are each triangle's **hypotenuse** and are equal

$DE = MO$. so, triangle DEF and MNO are congruent by RHS.

2) Triangle TUV is isosceles, where $TU = TV$.

TW is perpendicular to UV .

Prove triangles TUW and TWV are congruent by RHS.

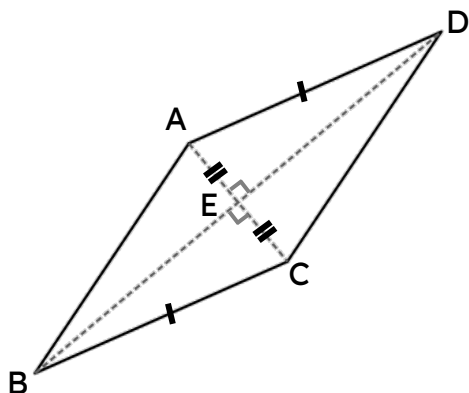


$\angle TWU = \angle TWV$ as TW is perpendicular to UV .

$TU = TV$ as triangle TUV is isosceles.

TW is a common side to both triangle TUW and TWV . so, triangles TUW and TWV are congruent by RHS.

3) By using known facts about rhombi and their diagonals prove that triangle AED is congruent to BEC by RHS.



$\angle AED = \angle BEC = 90^\circ$ as the diagonals of a rhombus are perpendicular.

$AD = BC$ as they are edges of a rhombus.

$AE = EC$ as the diagonals of a rhombus bisect each other.

So, triangle AED and BEC are congruent by RHS.

